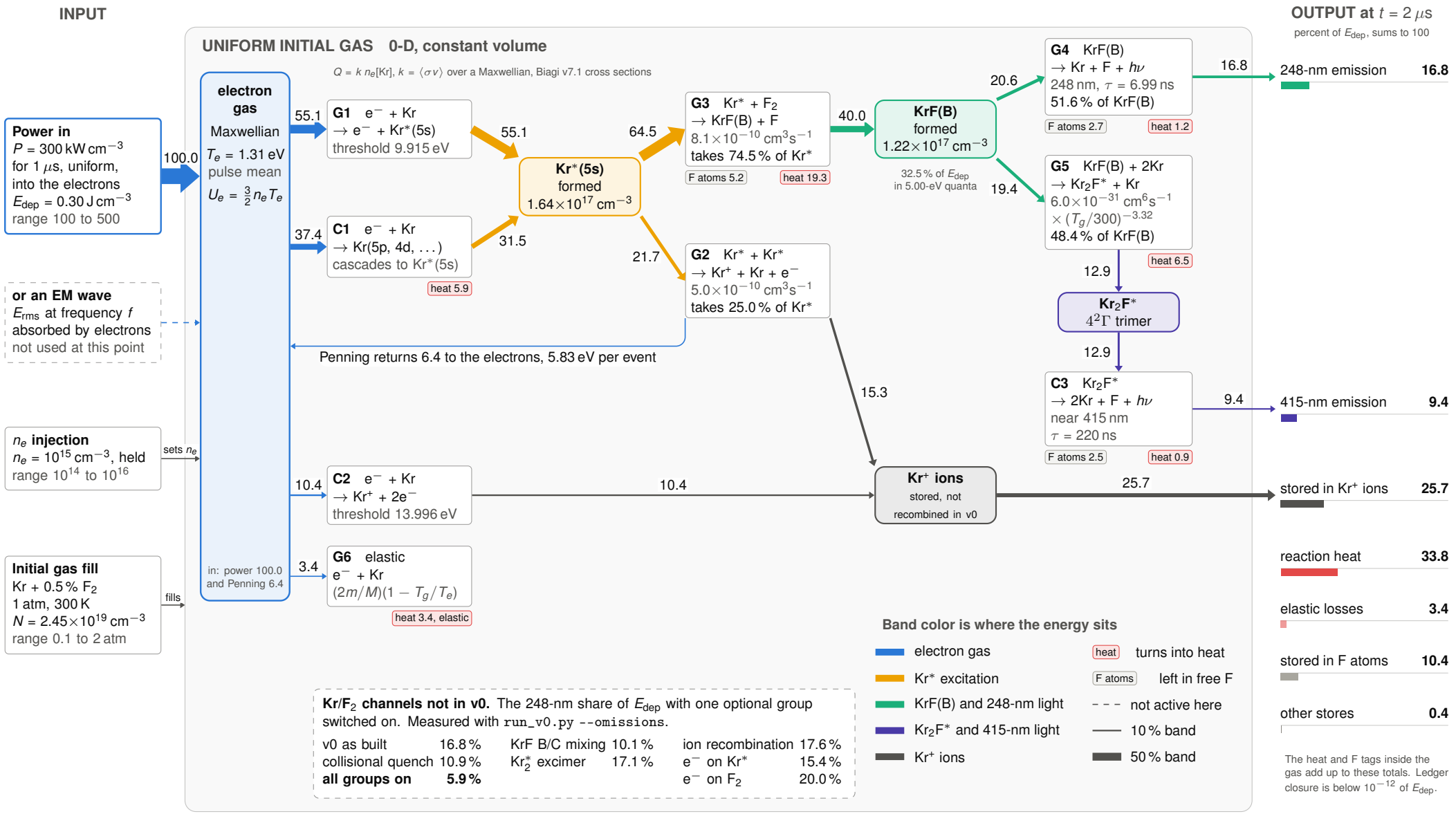
LIGHT v0 system map: energy flow through the Kr/F_2 kinetics

Reference point: 1 atm of Kr with 0.5 % F_2 at 300 K, P = 300 kW cm^{-3} for 1 μs so that E_{dep} = 0.30 J cm^{-3}, n_e = 10^{15} cm^{-3} held, run to 2 μs. Band width is proportional to the energy flow. Numbers are percent of E_{dep}, rounded to 0.1.



The notes' outputs as flows, which overlap: to Kr* 92.5 %, to Kr+ 25.7 %, to KrF(B) 32.5 %, heat 33.8 %, elastic losses 3.4 %, emission 26.2 %. At the end of the run T_g = 519 K and p = 1.74 atm. F_2 falls below 10 % of the fill at 830 ns.

Key to the map

How to read it. Each band carries energy. Its width is proportional to that energy, and its number is the energy as a percent of the deposited energy E_{dep}. A band's color is the store the energy sits in while it travels. Red tags are energy that turns into heat at that step. Grey tags are energy left behind in free fluorine atoms, because every harpoon event breaks an F_2 bond. Dashed outlines mark what is not active at the reference point.

Energies are counted at the formation energies below, measured from ground-state Kr and F_2 at rest. What enters a box therefore equals what leaves it. The script checks that for every box before it writes a number. Printed values are rounded to 0.1, so a box can be off by 0.1 between its inputs and its outputs.

The electrons get their energy from the power and from Penning ionization, which hands 5.83 eV per event back to them. They lose it to excitation, ionization and elastic collisions. Energy can therefore pass through the electrons more than once, and the notes' flows can add up to more than 100 %. Only the output ledger is a partition. Its last line, other stores, is what is still held in Kr^*, KrF(B), Kr_2F^* and the electron gas when the run stops.

Formation energies in eV above ground-state Kr and F_2

$Kr^*(5s)$	Kr^+	KrF(B)	Kr_2F^*	F
9.915	13.996	6.15	4.10	0.80

The photons carry 5.00 eV at 248 nm and 3.0 eV near 415 nm.

State at the reference point

T_e, pulse mean	1.31 eV
T_g and p at 2 μs	519 K, 1.74 atm
Kr^* formed	1.64×10^{17} cm^{-3}
KrF(B) formed	1.22×10^{17} cm^{-3}
F_2 below 10 % of the fill	at 830 ns
F_2 left at 2 μs	0.3 %

Regenerate after any change to the model:

```
cd LIGHT/docs
python3 make_system_map.py
latexmk -pdf system_map.tex
```

The v0 reactions. G-rows are the six in `KrF_model_notes_with_Gary.rtf`. C-rows are the closures that the notes' outputs need.

	reaction	rate	source
G1	$e^- + Kr \to e^- + Kr^*(5s)$	Maxwellian $\langle\sigma v\rangle$	Biagi v7.1, the whole 5s manifold at 9.915 eV
G2	$Kr^* + Kr^* \to Kr^+ + Kr + e^-$	5.0×10^{-10} cm^{3}s^{-1}	Kannari 1985, Penning products
G3	$Kr^* + F_2 \to KrF(B) + F$	8.1×10^{-10} cm^{3}s^{-1}	Kannari 1985 Table I
G4	$KrF(B) \to Kr + F + h\nu$, 248 nm	τ = 6.99 ns	Kannari 1985. The notes say about 4 ns
G5	$KrF(B) + Kr + Kr \to Kr_2F^* + Kr$	6.0×10^{-31} cm^{6}s$^{-1}\times(T_g/300)^{-3.32}$	Xu, Gadomski and Setser 1993, with Shui's T_g law
G6	$e^- + Kr$ elastic	power $\propto (2m/M)(1 - T_g/T_e)$	Biagi v7.1 momentum transfer
C1	$e^- + Kr \to Kr$(5p, 4d, higher), then Kr^*	Maxwellian $\langle\sigma v\rangle$	Biagi v7.1, prompt cascade to 5s
C2	$e^- + Kr \to Kr^+ + 2e^-$	Maxwellian $\langle\sigma v\rangle$	Biagi v7.1, 13.996 eV
C3	$Kr_2F^* \to 2Kr + F + h\nu$	τ = 220 ns	Xu, Gadomski and Setser 1993

Not in v0. The optional groups carry the rest of the pure Kr/F_2 chemistry in SPECK/REBUILD. The last column is the 248-nm share of E_{dep} at the reference point with that group switched on. It is 16.8 % in v0 as built.

group	what it adds	248 nm
quench	KrF(B) and Kr_2F^* quenched by Kr, F_2 and electrons	10.9 %
bc	the KrF C state, which mixes fast with B and radiates ten times slower	10.1 %
kr2	$Kr^* + 2Kr \to Kr_2^*$, its 147-nm light and its own harpoon	17.1 %
ions	$Kr^+ \to Kr_2^+$ and recombination back to Kr^*	17.6 %
electron	stepwise ionization and superelastic loss of Kr^*	15.4 %
f2e	electrons on F_2, meaning attachment, dissociation and vibration, and ion-ion recombination to KrF(B)	20.0 %
all	every group at once	5.9 %